

目的: 了解别人怎么放资料

取出别人放的数据 algorithm.

Ex: sort

map node and link

9/14 →

data type

signed	{	int	/	float
unsigned		long		double
		long long		
		short		char.

整数

data structures

- array
- linked list
- stack
- queue
- Tree
- Graph
- hash table

Array
Linked
List

知道头
都可往
后找

$[0] \sim [n-1]$ (index)

easy find 尾巴 info. (array 数固定的)

data type + array → struct { }

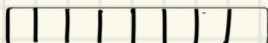
int → int[]

char → string (char[])

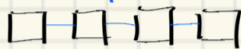
Linked List

把资料串起来 (靠 link)

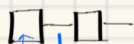
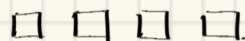
Array (自然连在一起)



① relationship

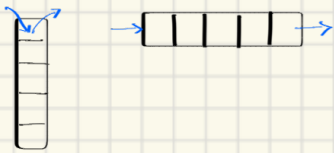


② no relationship



data → address.

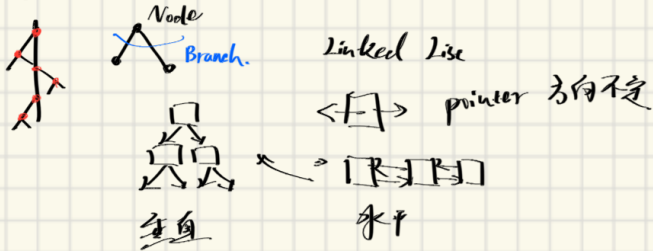
Stack / queue



access method difference

⇒ array - Linked List 完成 Stack / queue

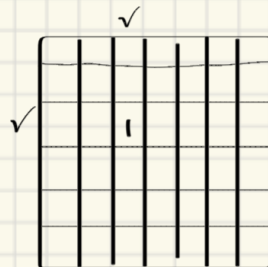
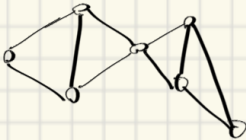
Tree



Graph

Vertex = V → V^2 array

edge = E



adjacent Matrix (array)

$V_1 \square \rightarrow \square \rightarrow \dots \rightarrow$ adjacent list
 $V_2 \square \rightarrow \square$

Basic concept

Array

struct $\begin{bmatrix} \text{address} \\ \text{data-type} \end{bmatrix} \rightarrow$ Linked List

Hash Table

hash function
data storage

array + Linked List
mod 5.

$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 \end{bmatrix} \leftarrow$ Table

↓
Data

↓
Data

Array → Linked list

insertion
deletion
be costly

Problem:

找 index 们
以后代码
很高

Stack

LIFO Lose in first out.